

Build an AI Interior Design Agent

Associate Product Manager — Product & Technology
Interior Company × Blocks

Time	About 6–8 hours of effort, spread across 3 days. Depth beats breadth.
What you build	A small but working AI agent, plus a lightweight harness that proves it works.
Tools	Any AI build tools, models, or APIs are welcome and encouraged.
Files with this brief	This document + interior_company_catalog.db (a SQLite database).

Confidential. Please do not share this brief, the dataset, or your solution publicly. The dataset is synthetic and for this exercise only.

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1. About this exercise

This is a build challenge, not a quiz. We want to see you design and ship a small but real AI product end to end: take a customer need, build something that works, and show how you know it works.

We are not grading pixel-perfect design or production-grade code. **We are looking at your product judgment, how well you direct AI tools, and whether you can prove your product is good enough to ship.**

Using AI build tools is expected and encouraged — that is how the role works. Tell us what you used and where you stepped in to correct it. Please keep effort to roughly 6–8 hours; a focused, working slice beats a sprawling, half-finished one.

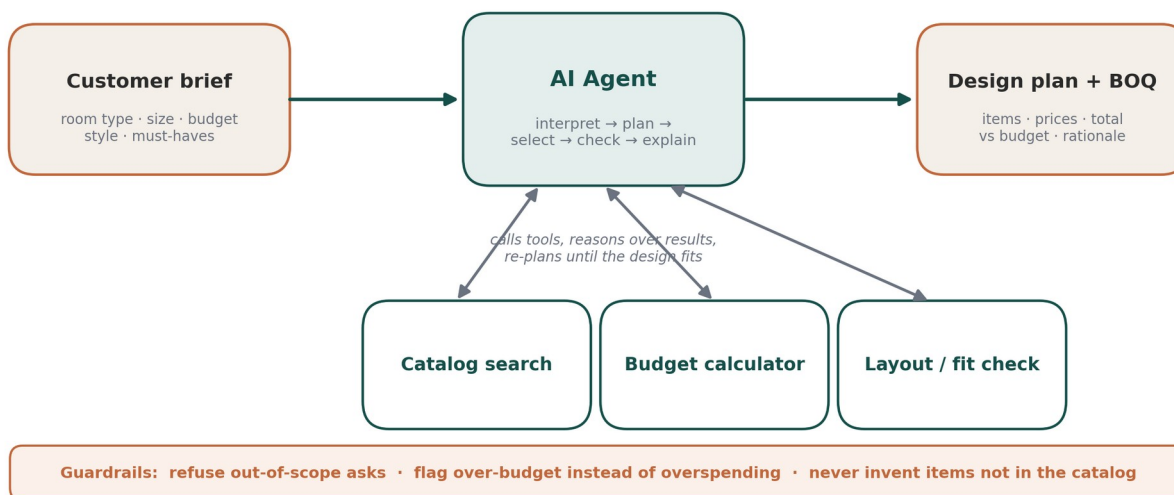
2. Who you would be building for

Interior Company designs full-home interiors for customers across India, and Blocks is the design platform our teams build on. As an Associate Product Manager here, you would own a product surface end to end, including the unglamorous but critical question every AI product eventually faces: how do we know it actually works? This exercise is a small version of exactly that job.

3. Your challenge at a glance

Build a working AI agent that turns a customer's room brief into a budget-fit design plan. The agent reads what the customer wants (room, size, budget, style, must-haves), uses tools to pick real products from our catalog, keeps the plan within budget and within the room, explains its choices — and knows what it should *not* do.

How the agent works



4. The scenario

A customer comes to us with a brief like the one below. They do not want a mood board of things they cannot buy. They want a plan they can act on: real products, within their budget, that fit their room, with a clear reason for each choice.

Example brief (BR-01)	Details
Room	Living room, 4.8 m × 3.6 m, 3.0 m ceiling
Budget	₹2,50,000
Style	Scandinavian
Must-haves	3-seater sofa, coffee table, TV unit, rug, lighting
Context	South-facing, lots of light; couple, no kids yet; wants a calm, bright room

Every brief in this format is already in the database (table `room_briefs`). You can design for those, accept free-text briefs, or both.

5. What to build (functional requirements)

Given a room brief, your agent should:

- Interpret the brief — room type, size, budget, style, and must-haves.
- Select products from our catalog that fit the style and the space.
- Stay within budget, and track how much is left as it builds the plan.

- Respect the room — the chosen pieces should physically fit and leave room to move.
- Explain its choices — a short rationale for the design and key picks.
- Be honest about trade-offs — what it prioritised, and what it left out and why.

The output is a design plan plus an itemised **Bill of Quantities (BOQ)**: each item with its price, the running total against budget, and what remains. A customer should be able to read it and act on it.

6. Make it agentic, not a chatbot

This must be an **agent**: it uses tools and reasons over several steps, rather than answering in one shot from the model's head. Build **all three** of the working tools below. The agent should call them, read the results, and re-plan until the design fits. The layout / fit check can be a simple, sensible heuristic — we want to see the agent reason about space, not a CAD engine.

Tool	What it does
Catalog search	Query the database for items by category, style, price range, and dimensions.
Budget calculator	Sum selected items, compare to the budget, and report what is left.
Layout / fit check	Check that selected pieces fit the room footprint with sensible circulation space. A simple, sensible heuristic is fine.

A chatbot would...	Your agent should...
Answer from memory and make up plausible products	Search the real catalog and only use items that exist in it
Confidently exceed the budget	Track spend with a tool and stop, or flag, when it runs out
Suggest pieces that cannot fit the room	Check dimensions before committing to a piece
Give one fixed answer	Re-plan when a tool result rules out its first choice

7. The data we give you (SQLite)

Alongside this brief you have **interior_company_catalog.db**, a SQLite database with two tables. It is the world your agent lives in.

catalog — 72 furniture and decor items. **This is the only source of truth for products. Your agent must never invent items that are not in this table.**

Column	Meaning
item_id	Unique product code (e.g. SOF-001)
category	Sofa, Bed, Dining Table, Rug, Floor Lamp, etc.
name	Product name
style_tags	Comma-separated styles (Scandinavian, Mid-Century, Industrial, ...)
price_inr	Price in ₹. May be NULL where price is not yet loaded.
width_cm, depth_cm, height_cm	Dimensions in cm. May be NULL for some items.
color_finish	Colour / material finish
in_stock	1 = in stock, 0 = out of stock
lead_time_days	Days to deliver
room_types	Comma-separated rooms it suits (Living Room, Bedroom, ...)

room_briefs — 14 sample customer briefs to design for and test against.

Column	Meaning
brief_id	Brief code (e.g. BR-01)
room_type	Living Room, Bedroom, Dining, Study, Kids
length_cm, width_cm, ceiling_cm	Room dimensions in cm
budget_inr	Customer budget in ₹
style_preference	Requested style
must_haves	Pieces the customer specifically wants
constraints	Practical constraints (rented, kids, orientation, ...)
customer_note	The request in the customer's own words

The mess is intentional. Some items have missing prices or dimensions, some are out of stock with long lead times, and a few briefs are deliberately tricky. Real product data looks like this. Your agent — and your evals — should handle it gracefully rather than break or pretend.

A couple of example queries to get started:

```
SELECT * FROM catalog WHERE category='Sofa' AND price_inr <= 60000 AND in_stock=1;
SELECT * FROM catalog WHERE style_tags LIKE '%Scandinavian%' AND room_types LIKE '%Living Room%';
```

8. Guardrails: what the agent must not do

A good product knows its limits. Your agent should:

- Never recommend products that are not in the catalog, even if the customer names a specific brand or designer piece.
- Never silently go over budget. If the must-haves do not fit the budget, say so and offer trade-offs.
- Decline out-of-scope requests. Civil, structural, electrical, or plumbing advice (for example, “should I remove this wall?”) is not ours to give — point the customer to a qualified human expert.
- Not promise guaranteed delivery dates or final negotiated prices.
- Be honest when a request is not possible (budget too low, room too small) and offer the closest realistic option instead of faking a plan.

Several briefs in the dataset will test exactly these situations. Finding them is part of the exercise.

9. Prove it works: the evaluation harness

This is the most important part of the exercise, and the part most candidates underinvest in. Anyone can demo a happy path. We want to see that you can tell, with evidence, whether your agent is good enough to ship.

Build a lightweight eval harness that includes:

- A golden set of 20–30 test cases. Start from the briefs in the database, then add adversarial cases and at least a few “we cannot do this” cases.
- A mix of scorers. Deterministic checks the code can verify (Is the total within budget? Are all chosen items in the catalog? Do the pieces fit the room? Did it correctly decline the out-of-scope brief?), plus at least one judgement-based scorer for something subjective (style coherence or quality of the explanation), for example an LLM-as-judge with a written rubric.
- Behaviour and tool use, not just final text. Check what the agent did (did it actually consult the budget tool? did it avoid recommending an out-of-stock item by accident?), not only how the answer reads.
- A ship gate. State the thresholds at which you would call it good enough to ship (for example: budget never exceeded, 100% of items real, style rated acceptable on at least 90% of cases).
- Honest results. Run the harness and report what you find, including the cases that fail. Failures you have found and understood are a strong signal, not a weakness.

10. Ground rules and scope

Time: About 6–8 hours across 3 days. We read the core of your submission most closely, so prioritise.

Scope: One room done well beats a whole home done shallowly. Pick a single room type and go deep.

Tools: Any AI build tools, models, or APIs are welcome (for example: Lovable, v0, Cursor, Bolt, Replit, Claude Code). Tell us what you used.

Skip the polish: Do not spend time on sign-in, multiple users, pixel-perfect UI, or fancy deployment. A clean, functional demo is enough.

11. What to submit

Deliverable	What we expect
A. A runnable MVP	A deployed link, or a repo with clear run steps. We should be able to run it in under five minutes.
B. The eval harness	Your golden set, your scorers, and a short results report that shows the failures too.
C. A decision log	One page or a 5-minute video: what you scoped in and out and why; how you directed the AI tools and where you overrode them; what would break in production; and what you would build next.

12. How we will evaluate it

We look at four things in roughly equal measure. The decision log and the live conversation matter as much as the running app, because they reveal the thinking the demo cannot.

What we look at	What a strong submission shows
Product judgment & ownership	A sharp, well-scoped slice; clear non-goals; sensible trade-offs you can defend; decisions made independently.
Agentic engineering	Real, working tool use and multi-step reasoning; graceful handling of edge cases; sensible guardrails — not a thin wrapper.
Evaluation rigor	A real golden set, a mix of deterministic and judgement scorers, tool-use checks, a ship gate, and honest results.
Directing AI tools	Evidence that you steered the tools well, caught their mistakes, and used your taste — not just accepted the first output.

13. The follow-up conversation

If your submission moves forward, we will invite you to a 25-minute call. You will give a quick demo, walk us through one of your eval cases and why it tests what you claim, and make a small change live (for example, a new constraint we hand you on the spot). It is worth building your solution so you can stand behind every decision in it.

14. A few tips

- Read three or four briefs end to end before you build. The trickiest ones will shape your design.
- Get one brief working all the way through before you widen scope.
- Treat your evals as a product spec, not an afterthought. Writing the test cases will sharpen the agent.
- When in doubt, make the honest choice (flag, decline, ask) rather than the impressive-looking one.
- State your assumptions in the decision log. We would rather see a clear assumption than a guess dressed up as fact.

15. Submission and timeline

Please reply to this email with your submission **within 3 days** of receiving this brief. If anything is unclear, simply reply on email with your questions and we will get back to you.

A suggested way to spend the time (entirely optional):

Day	Focus
Day 1	Read the briefs and the data. Scope your slice. Get one brief working with catalog search and a basic plan.
Day 2	Add the budget and layout tools and the guardrails. Build your golden set and first scorers.
Day 3	Run and tidy the evals, write the decision log, and package the demo. Stop at the time cap.

We are genuinely looking forward to seeing how you think. Have fun with it.